

A GAUSSIAN-KERNEL REDUCTION OF SMALL-UNCERTAINTY DISCRIMINATORY LEARNING

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ABSTRACT

We derive and implement a controlled small-uncertainty reduction of the microscopic entropic learning dynamics used in the discriminatory society model. The reduction keeps the original modulation functions, rescales time by the small uncertainty scale, and averages only over Gaussian issue vectors. The result is a set of low-dimensional pairwise kernels depending on ideological overlap, distrust, and the discrimination field. We then close these kernels at the population level using four class/status prototypes and obtain a semi-analytical phase diagram over discrimination strength d and discriminator fraction f_d . The main signal is affective: the class-trust order parameter $R_{\mu,c}$ changes sign with d and grows in magnitude with f_d , while the ideological class-alignment order parameter $R_{c,w}$ is essentially constant in this closure.

1 SMALL- C, V REDUCTION

For receiver r , emitter e , and issue vector x , the microscopic fields are

$$h_w = \frac{\sigma_e \hat{w}_r^\top x}{\sqrt{1 + x^\top C_r x}}, \quad h_\mu = \frac{\mu_{e|r}}{\sqrt{1 + V_{e|r}}}, \quad \sigma_e = \text{sign}(\hat{w}_e^\top x). \quad (1)$$

Discrimination enters by shifting the ideological field, $H = h_w + D_{e|r}$. The evidence and modulation functions are kept exactly as in the microscopic model:

$$Z(H, \mu) = \Phi(H) + \Phi(\mu) - 2\Phi(H)\Phi(\mu), \quad (2)$$

$$F_w(H, \mu) = \frac{[1 - 2\Phi(\mu)]\phi(H)}{Z(H, \mu)}, \quad F_\mu(H, \mu) = \frac{[1 - 2\Phi(H)]\phi(\mu)}{Z(H, \mu)}, \quad (3)$$

$$F_C(H, \mu) = -F_w(H, \mu)[F_w(H, \mu) + H], \quad F_V(H, \mu) = -F_\mu(H, \mu)[F_\mu(H, \mu) + \mu]. \quad (4)$$

Writing $C_r = \epsilon \bar{C}_r$ and $V_{e|r} = \epsilon \bar{V}_{e|r}$ with $\epsilon \ll 1$, the nontrivial limit is obtained on slow time $\tau = \epsilon t$. To leading order,

$$\frac{d\hat{w}_r}{d\tau} = \bar{C}_r x \sigma_e F_w(h + D_{e|r}, \mu_{e|r}), \quad (5)$$

$$\frac{d\mu_{e|r}}{d\tau} = \bar{V}_{e|r} F_\mu(h + D_{e|r}, \mu_{e|r}), \quad (6)$$

where $h = \sigma_e \hat{w}_r^\top x$. The corresponding equations for $\bar{C}_r$ and $\bar{V}_{e|r}$ use F_C and F_V with the same leading fields.

2 GAUSSIAN PAIRWISE KERNELS

Let $u_r = \hat{w}_r^\top x$ and $u_e = \hat{w}_e^\top x$. For isotropic Gaussian issues, (u_r, u_e) are jointly Gaussian. The microscopic ideological field $h = \text{sign}(u_e)u_r$ has density

$$p(h | q_r, \rho) = \frac{2}{\sqrt{q_r}} \phi\left(\frac{h}{\sqrt{q_r}}\right) \Phi\left(\frac{\rho h}{\sqrt{q_r(1 - \rho^2)}}\right), \quad (7)$$

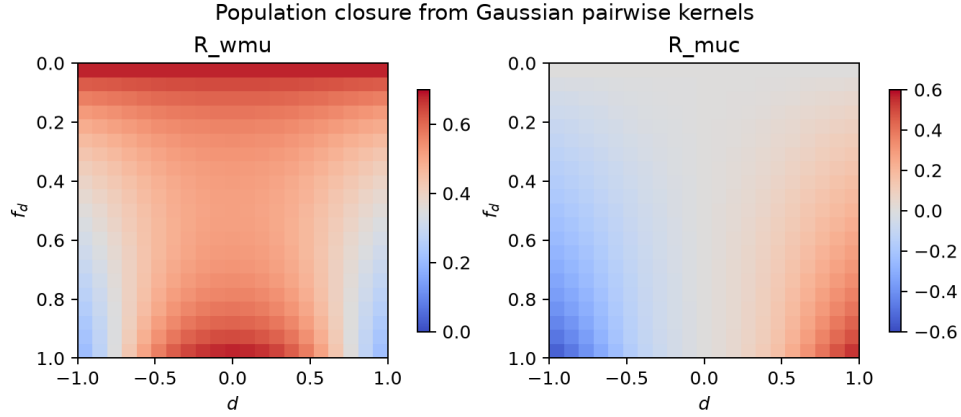


Figure 1: Semi-analytical population phase diagram from the Gaussian-kernel closure. Left: $R_{w,\mu}$, plotted on the range $[0, 0.7]$. Right: $R_{\mu,c}$, plotted on the range $[-0.6, 0.6]$. The ideological class-alignment observable $R_{c,w}$ is not shown because it is nearly constant across this sweep ($R_{c,w} \simeq 6.3 \times 10^{-3}$).

with separate one-sided limits as $\rho \rightarrow \pm 1$. This gives the affective kernels

$$\mathcal{M}_\mu(q_r, \rho, \mu, D) = \int p(h | q_r, \rho) F_\mu(h + D, \mu) dh, \quad (8)$$

$$\mathcal{M}_V(q_r, \rho, \mu, D) = \int p(h | q_r, \rho) F_V(h + D, \mu) dh. \quad (9)$$

The ideological drift is written as

$$\mathbb{E}_x[x \text{ sign}(u_e) F_w(h + D, \mu)] = A(q_r, q_e, \rho, \mu, D) \hat{w}_r + B(q_r, q_e, \rho, \mu, D) \hat{w}_e, \quad (10)$$

where A and B are computed from Gaussian moments. The implementation uses Gauss–Hermite quadrature and was checked against direct Monte Carlo estimates of the original Gaussian issue averages.

3 POPULATION CLOSURE

To obtain a population-level phase diagram, we group agents into four prototypes: class A/B crossed with nondiscriminatory/discriminatory receiver status. The population weights are

$$\frac{1 - f_d}{2}, \quad \frac{f_d}{2}, \quad \frac{1 - f_d}{2}, \quad \frac{f_d}{2}.$$

For the discriminatory receivers we use the symmetric case-6 field

$$D = d \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix},$$

and nondiscriminatory receivers have $D = 0$. For each ordered prototype pair, the affective variables evolve according to

$$\dot{\mu}_{e|r} = \bar{V}_{e|r} \mathcal{M}_\mu(1, \rho_{re}, \mu_{e|r}, D_{e|r}), \quad \dot{\bar{V}}_{e|r} = \bar{V}_{e|r}^2 \mathcal{M}_V(1, \rho_{re}, \mu_{e|r}, D_{e|r}),$$

with emitter-rate weights given by the prototype proportions. Ideological prototype vectors are evolved using the A, B drift and then renormalized.

4 RESULT

Figure 1 is the main output of the reduction. The closure predicts a robust affective ordering: for $d > 0$, discriminatory receivers become more trusting of their own class and less trusting of the

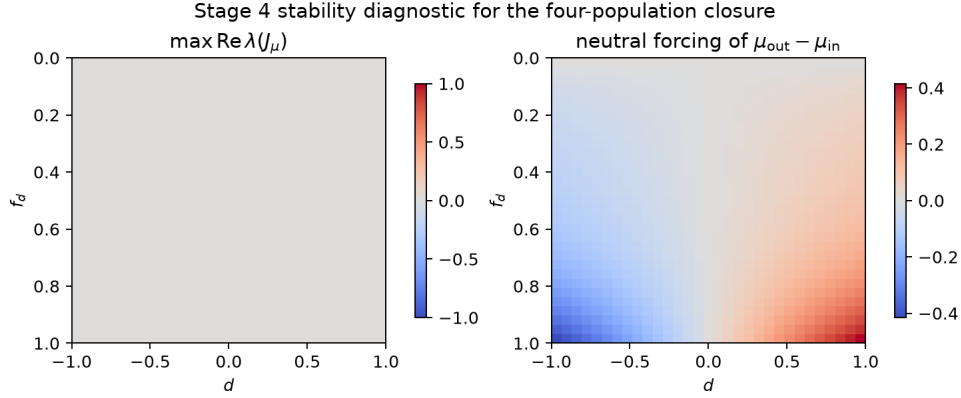


Figure 2: Stage-4 diagnostic for the current four-population closure. Left: the largest real eigenvalue of the homogeneous affective block J_μ , which remains zero across the grid. Right: the direct neutral forcing of the class-trust contrast $\mu_{\text{out}} - \mu_{\text{in}}$, which changes sign with d and grows with f_d . In this closure the class-trust phase diagram is therefore a driven response rather than an eigenvalue-crossing instability.

other class, giving positive $R_{\mu,c}$; reversing the discrimination field reverses the sign. The magnitude of this affective order increases with f_d . The mixed trust-ideology observable $R_{w,\mu}$ remains positive over the full grid and is largest when the discriminatory fraction is high and $|d|$ is not too large. In contrast, $R_{c,w}$ is essentially flat in the present closure. This means that the reduced pairwise kernels already capture class-structured trust, but the current low-rank population closure does not generate a strong ideological class split.

5 STAGE 4: NEUTRAL-STATE STABILITY

The rebuild plan calls for a neutral-state stability calculation and, ideally, a critical curve $d_c(f_d)$. We tested this within the four-population closure by linearizing the affective subsystem around the class-symmetric neutral configuration. For $d \neq 0$, however, the neutral state is not an unforced fixed point: the discrimination field enters the pairwise kernel directly and creates an additive class-contrast forcing. Thus the linearized class-contrast equation has the schematic form

$$\frac{d}{d\tau}(\mu_{\text{out}} - \mu_{\text{in}}) = J_\mu(\mu_{\text{out}} - \mu_{\text{in}}) + b(f_d, d),$$

where $b(f_d, d)$ is nonzero whenever discriminatory receivers are present and $d \neq 0$.

Figure 2 shows the outcome. The homogeneous affective block does not produce a positive eigenvalue or a zero-crossing curve in this approximation. Instead, the same structure visible in $R_{\mu,c}$ is already present as a direct forcing term at the neutral state. Consequently, this closure does not yet deliver the desired theoretical transition line $d_c(f_d)$. It does identify the source of the class-trust ordering as the sign-structured pairwise kernel forcing induced by $D_{e|r}$.

6 TWO-CLASS ATTRACTION BOUNDARY

We also examined a simpler two-class reduction that gives a cleaner analytical condition for ideological attraction versus repulsion. Suppose each class is represented by one normalized prototype vector and let ρ denote the cross-class overlap. For a cross-class interaction, the tangent part of the ideological drift reduces to

$$\dot{\rho} = c_{\text{eff}} B(\rho, \mu_{\text{out}}, D_{\text{out}})(1 - \rho^2),$$

where $c_{\text{eff}} > 0$ absorbs the interaction rate and the small uncertainty scale. Thus the sign of B determines whether the classes are locally pulled together or pushed apart in ideology. Closing the affective variable by the stationary pairwise condition

$$\mathcal{M}_\mu(\rho, \mu_{\text{out}}^*, -d) = 0$$

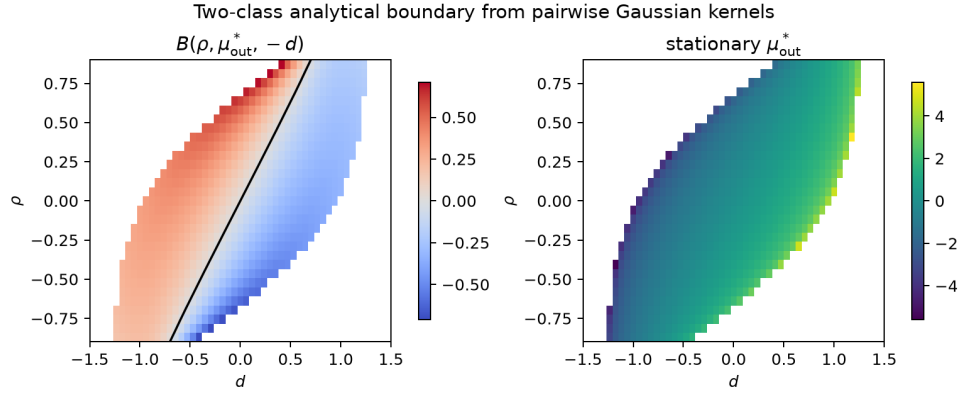


Figure 3: Two-class analytical attraction boundary. The black contour is $B(\rho, \mu_{\text{out}}^*, -d) = 0$. On one side of the contour the cross-class ideological drift is attractive, and on the other it is repulsive. White regions indicate parameter values where the stationary equation $\mathcal{M}_\mu = 0$ had no finite root in the search window.

gives the candidate boundary

$$B(\rho, \mu_{\text{out}}^*(\rho, d), -d) = 0.$$

Figure 3 is the most promising analytical boundary obtained so far. It is not yet the full $d_c(f_d)$ curve requested by the rebuild plan because f_d has been absorbed into the effective interaction rate in this two-class limit. Nevertheless, it gives an explicit kernel-level condition for when the ideological part of the dynamics changes from cross-class attraction to cross-class repulsion.

7 LIMITATIONS

This calculation is semi-analytical rather than fully analytical: Gaussian quadrature is used for the pairwise kernels, and the final population equation is a four-prototype closure. The nearly constant $R_{c,w}$ should therefore be read as a limitation of this closure, not as a proof that ideological ordering is absent in the microscopic model. A sharper closure should evolve the covariance kernel $\bar{C}_r$ rather than replacing each population by a normalized prototype vector. The Stage-4 calculation also shows that this closure does not yet supply a spontaneous neutral-state instability or a predicted $d_c(f_d)$. Obtaining such a line likely requires a closure in variables ρ_{ab} , $\mu_{b|a}$, and anisotropic uncertainty modes, followed by a Jacobian analysis of a true neutral fixed point.